

Open Science Practices for Demographic Research

Jonas Schöley

schoeley@demogr.mpg.de

 [@jschoeley.com](https://twitter.com/jschoeley)

jschoeley.com

Aliakbar Akbaritabar

akbaritabar@demogr.mpg.de

 [@akbaritabar.bsky.social](https://bsky.app/profile/akbaritabar.bsky.social)

akbaritabar.github.io



MAX PLANCK INSTITUTE
FOR DEMOGRAPHIC RESEARCH

Benefits of reproducibility and open science

Some personal experiences

Together with



Roland Rau
@Demographie

CRAN Task View: Survival Analysis

Maintainer: Arthur Allignol, Aurelien Latouche
Contact: arthur.allignol at gmail.com
Version: 2022-03-07
URL: <https://CRAN.R-project.org/view=Survival>
Source: <https://github.com/cran-task-views/Survival/>

Testing

- The `survdif` function in [survival](#) compares survival curves using the Fleming-Harrington G-rho family of test. [NADA](#) implements this class of tests for left-censored data.
- The [maxcombo](#) package compares survival curves using the max-combo test, which is often based on the Fleming-Harrington G-rho family of tests and is designed to have higher power than the logrank test in the scenario of non-proportional hazards such as those resulting from delayed treatment effects.
- [clinfun](#) implements a permutation version of the logrank test and a version of the logrank that adjusts for covariates.
- The [exactRankTests](#) implements the shift-algorithm by Streitberg and Roehmel for computing exact conditional p-values and quantiles, possibly for censored data.
- `SurvTest` in the [coin](#) package implements the logrank test reformulated as a linear rank test.
- The [maxstat](#) package performs tests using maximally selected rank statistics.
- The [interval](#) package implements logrank and Wilcoxon type tests for interval-censored data.
- Three generalised logrank tests and a score test for interval-censored data are implemented in the [glrt \(archived\)](#) package.
- [survcomp](#) compares 2 hazard ratios.
- The [TSHRC](#) implements a two stage procedure for comparing hazard functions.
- The [FHtest](#) package offers several tests based on the Fleming-Harrington class for comparing survival curves with right- and interval-censored data.
- The [LogrankA \(archived\)](#) package provides a logrank test for which aggregated data can be used as input.
- The short term and long term hazard ratio model for two samples survival data can be found in the [YPmodel](#) package.
- The [controlTest](#) implements a nonparametric two-sample procedure for comparing the median survival time.
- The [survRM2](#) package performs two-sample comparison of the restricted mean survival time
- The [emplik2](#) package permits to compare two samples with censored data using empirical likelihood ratio tests.
- The [KONPsurv](#) package provides powerful nonparametric K-sample tests for right-censored data. The tests are consistent against any differences between the hazard functions of the groups.

Together with



Ilya Kashnitsky
@ikashnitsky

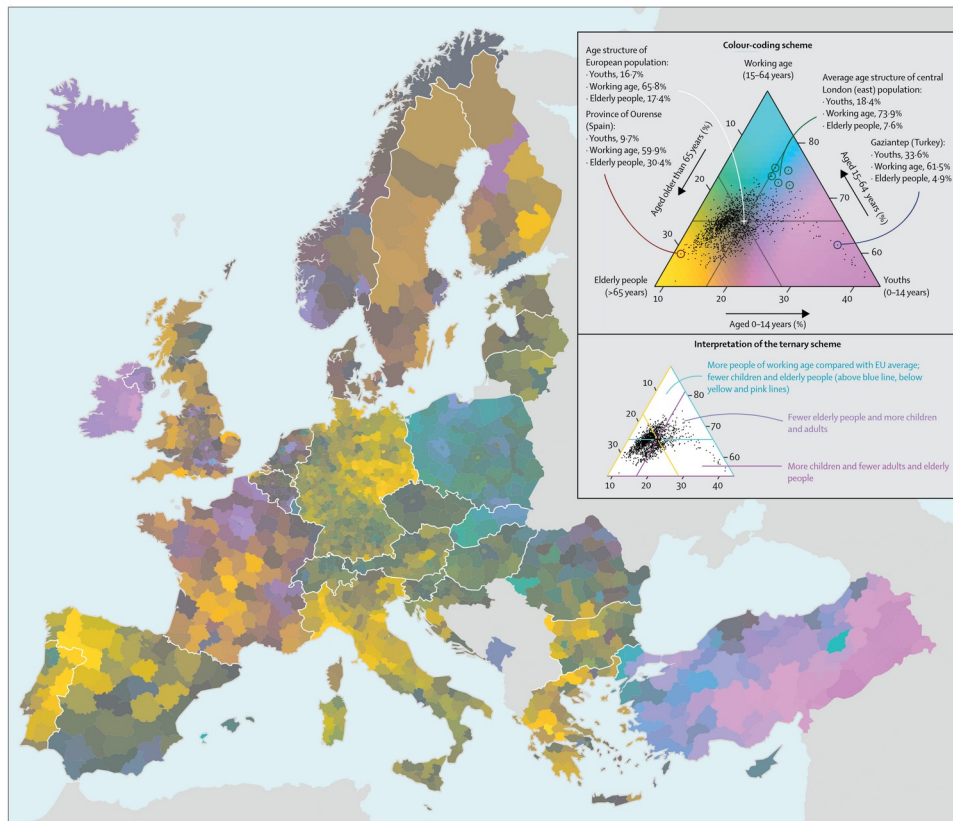


Figure: Colour-coded map of population structures in European Nomenclature of Territorial Units for Statistics 3 regions in 2015
Each population composition in the European Nomenclature of Territorial Units for Statistics 3 regions is uniquely colour coded. Colours show direction and magnitude of deviations from the centrepiece, which represents the average age of the European population, and is dark grey. The hue component of a colour encodes the direction of deviation: yellow indicates an elderly population (>65 years), cyan indicates people of working age (15-64 years), and magenta indicates children (0-14 years). Chroma and lightness components signify the distance from the centre ranging from desaturated and dark colours near the centre to vivid and bright colours at the corners. We provide R code to fully reproduce this map.²⁰

Kashnitsky & Schöley (2018). Regional population structures at a glance. [10.1016/S0140-6736\(18\)31194-2](https://doi.org/10.1016/S0140-6736(18)31194-2)

Together with



Ilya Kashnitsky
@ikashnitsky

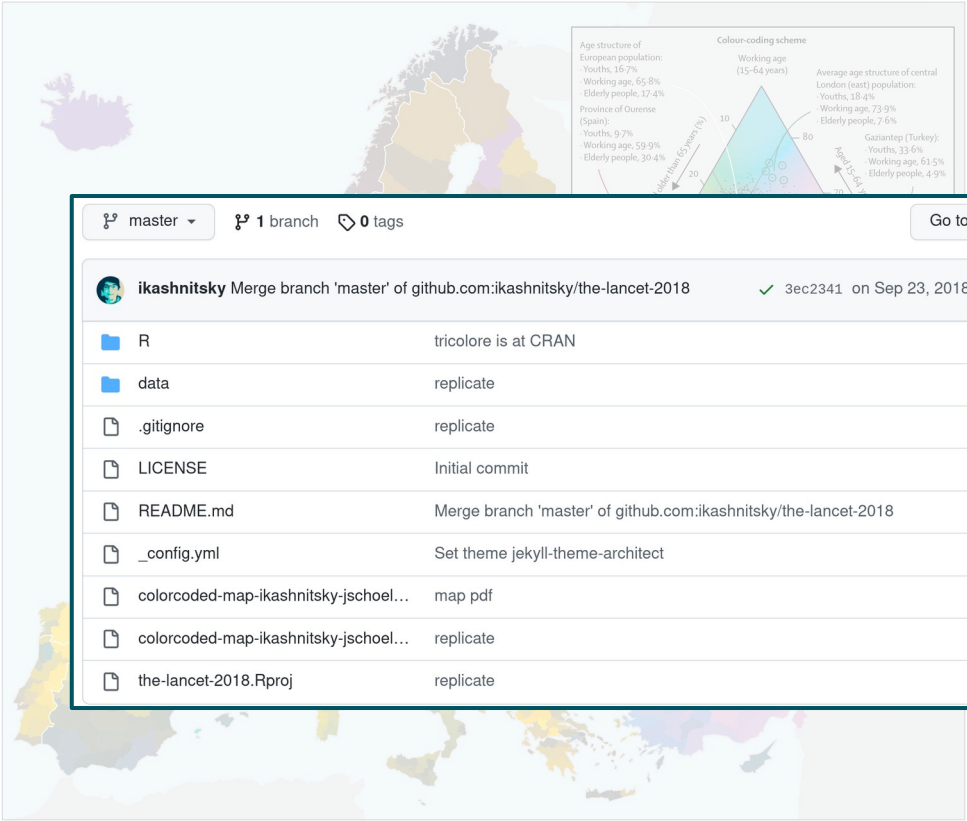


Figure: Colour-coded map of population structures in European Nomenclature of Territorial Units for Statistics 3 regions in 2015. Each population composition in the European Nomenclature of Territorial Units for Statistics 3 regions is uniquely colour coded. Colours show direction and magnitude of deviations from the centrepoint, which represents the average age of the European population, and is dark grey. The hue component of a colour encodes the direction of deviation: yellow indicates an elderly population (>65 years), cyan indicates people of working age (15–64 years), and magenta indicates children (0–14 years). Chroma and lightness components signify the distance from the centre ranging from desaturated and dark colours near the centre to vivid and bright colours at the corners. We provide R code to fully reproduce this map.²⁶

master 1 branch 0 tags

Go to file Code

About

ikashnitsky Merge branch 'master' of github.com:ikashnitsky/the-lancet-2018 3ec2341 on Sep 23, 2018 15 commits

R	tricolore is at CRAN	4 years ago
data	replicate	4 years ago
.gitignore	replicate	4 years ago
LICENSE	Initial commit	4 years ago
README.md	Merge branch 'master' of github.com:ikashnitsky/the-lancet-2018	4 years ago
_config.yml	Set theme jekyll-theme-architect	4 years ago
colorcoded-map-ikashnitsky-jschoel...	map pdf	4 years ago
colorcoded-map-ikashnitsky-jschoel...	replicate	4 years ago
the-lancet-2018.Rproj	replicate	4 years ago

Reproduce "Regional population structures at a glance"

[doi.org/10.1016/S0140-6736\(18\)31194-2](https://doi.org/10.1016/S0140-6736(18)31194-2)

Readme

MIT license

74 stars

7 watching

18 forks

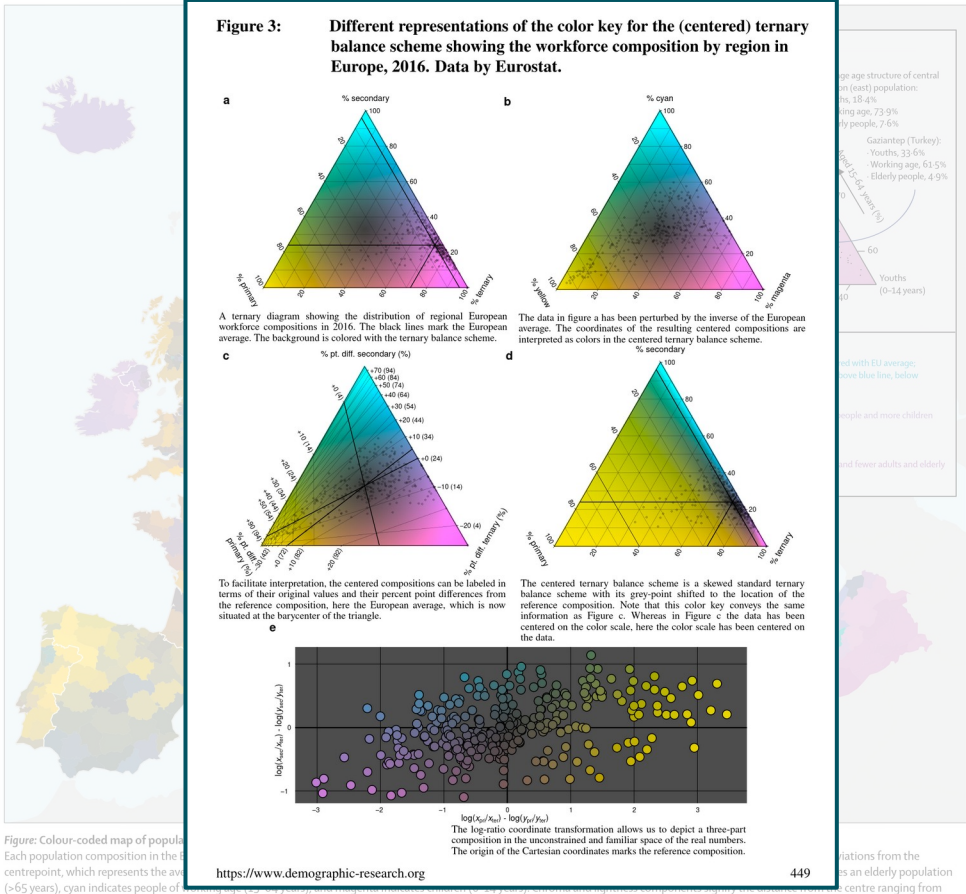
Releases

No releases published

Together with



Ilya Kashnitsky
@ikashnitsky



Together with



Ilya Kashnitsky
@ikashnitsky

tricolore. A flexible color scale for ternary compositions

Jonas Schöley & Ilya Kashnitsky

CRAN 1.2.0 build passing License GPL v3



What is *tricolore*?

tricolore is an R library providing a flexible color scale for the visualization of three-part (ternary) compositions. Its main functionality is to color-code any ternary composition as a mixture of three primary colours and to draw a suitable color-key. *tricolore* flexibly adapts to different visualization challenges via

- *discrete* and *continuous* color support,
- support for unbalanced compositional data via *centering*,
- support for data with very narrow range via *scaling*,
- *hue*, *chroma* and *lightness* options.

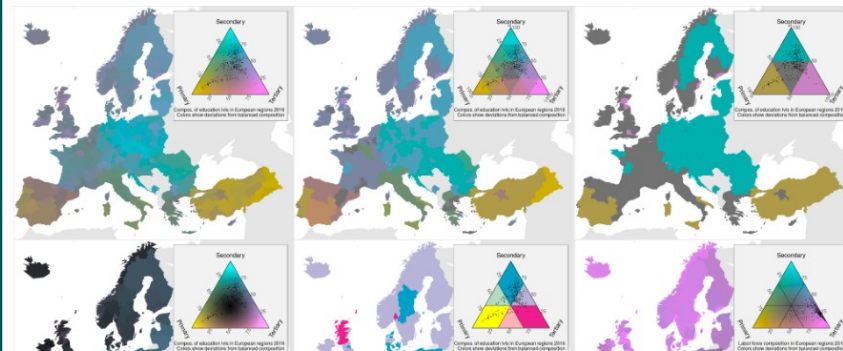
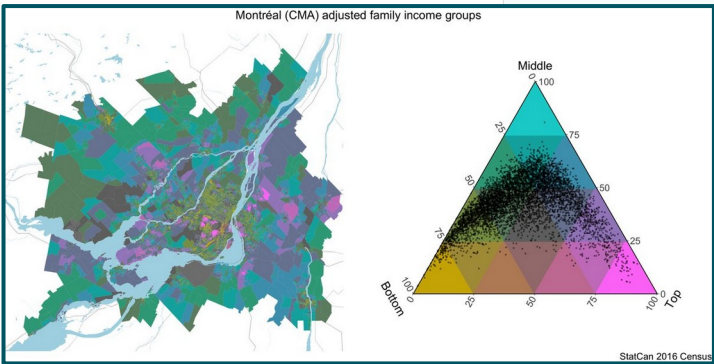


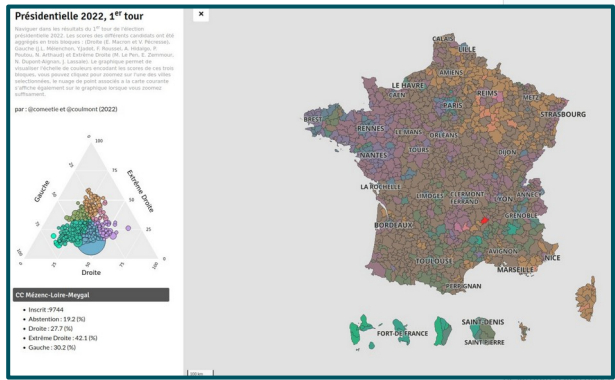
Figure: Colours of Europe
Each population composition in the European Nomenclature of territorial units for statistics 3 regions is uniquely colour coded. Colours show direction and magnitude of deviations from the centrepoint, which represents the average age of the European population, and is dark grey. The hue component of a colour encodes the direction of deviation: yellow indicates an elderly population (>65 years), cyan indicates people of working age (15–64 years), and magenta indicates children (0–14 years). Chroma and lightness components signify the distance from the centre ranging from desaturated and dark colours near the centre to vivid and bright colours at the corners. We provide R code to fully reproduce this map.¹⁶



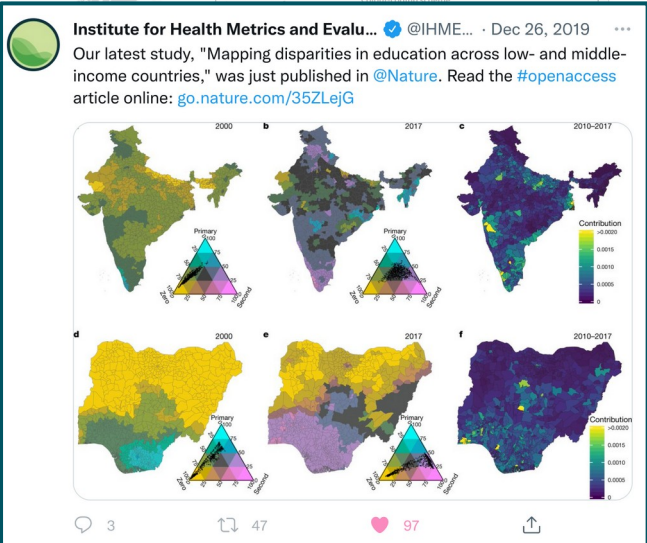
Income distribution in Canadian cities



French election results



LMIC education disparity



Vienna's population by origin

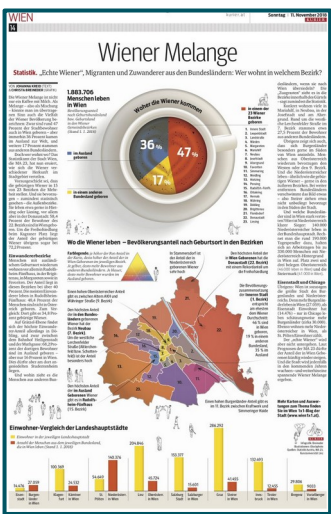
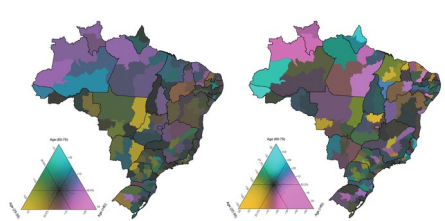


Figure 3: Spatial distribution of deaths from COVID-19 by age group in Brazilian meso-regions, males – first wave (left) and second wave (right)



Source: Ministry of Health (2021).

Regional age distribution of COVID deaths In Brazil

ations from the
an elderly population
centre ranging from

Agricultural and Forest Meteorology

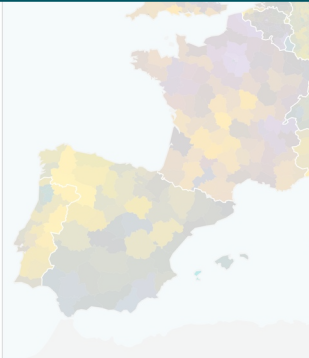
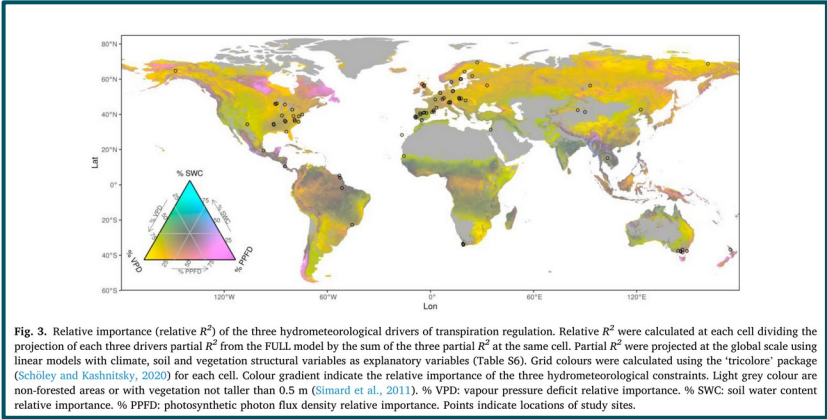
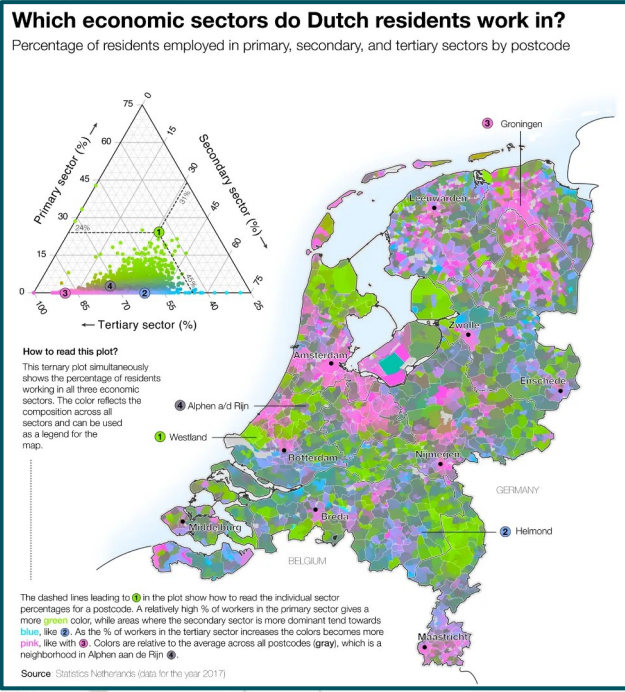
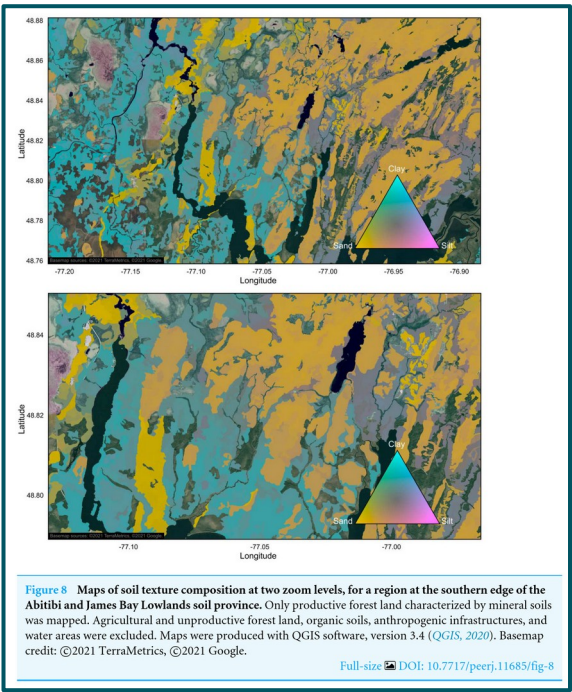


Figure: Colour-coded map of population structures in European Nomenclature of Territorial Units for Statistics 3 regions in 2015. Each population composition in the European Nomenclature of Territorial Units for Statistics 3 regions is uniquely colour coded. Colours show direction and magnitude of deviations from the centrepoint, which represents the average age of the European population, and is dark grey. The hue component of a colour encodes the direction of deviation: yellow indicates an elderly population (>65 years), cyan indicates people of working age (15–64 years), and magenta indicates children (0–14 years). Chroma and lightness components signify the distance from the centre ranging from desaturated and dark colours near the centre to vivid and bright colours at the corners. We provide R code to fully reproduce this map.⁸⁴

Labor force composition



Soil composition



Reproducible Demographers

Aliakbar **Akbaritabar** @akbaritabar.bsky.social Created a fully reproducible data base of academic migration flows.



Tom **Theile**

New version: SMD 2.0 is available since June 2026!

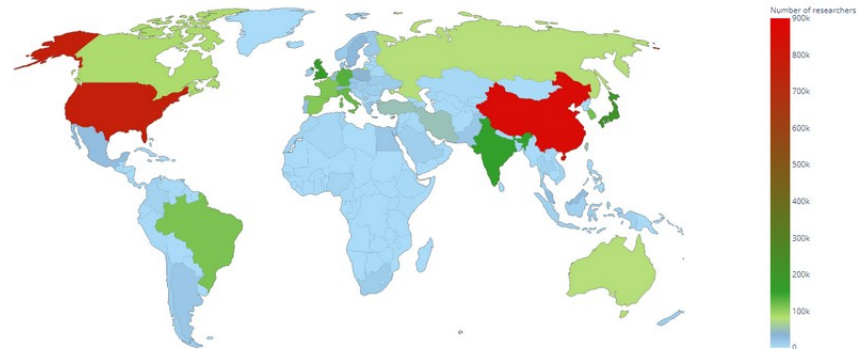
The new version is based on newer data (April 2026 for OpenAlex, October 2025 for Scopus) and adds more disaggregations: Gender, field of science and subnational location.

Go to the download section to find the new data downloads. We are updating the website during the next weeks with more information about the new data and new interactive visualizations. If you have any questions, please contact us via email.

> Research

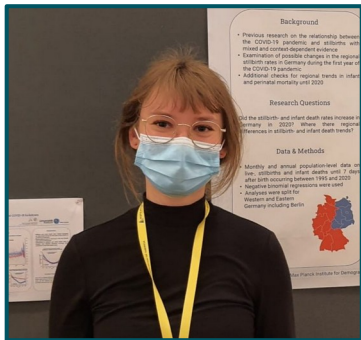
> How is the data produced?

Number of published scholars per country
mean number of scopus published researchers in the years 2013 - 2017



Reproducible Demographers

Tim **Riffe** @timriffe1 & Enrique **Acosta** @Acosta_Kike_ &
Manal **Elzalabany** & Maxi **Kniffka** @MaxiKniffka & Jessica **Donzowa** @jdonzowa

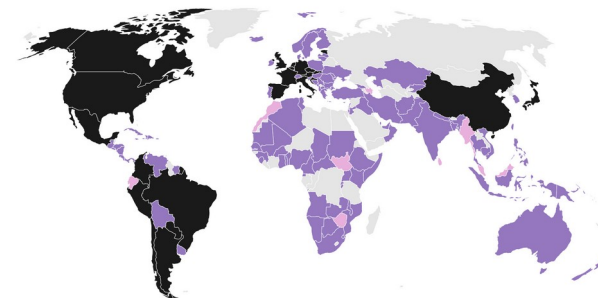


Created a fully reproducible data base of
age specific COVID-19 statistics.

Data availability

You can get the most up-to-date data at the [OSF](https://osf.io/mpwjg/) site that we mirror to: <https://osf.io/mpwjg/>.

Here's an overview of global coverage as of now. A country marked as *forthcoming* means we've identified a source, but that collection is pending for one reason or another. Are you from one of the countries not yet in the collection and want to pitch in? Are you interested in adopting collection for one of our time series that has fallen behind? We're in need of more support. Please reach out, if so by emailing us at coverage-db@demogr.mpg.de.



■ National and subnational ■ National ■ Forthcoming ■ Not included yet

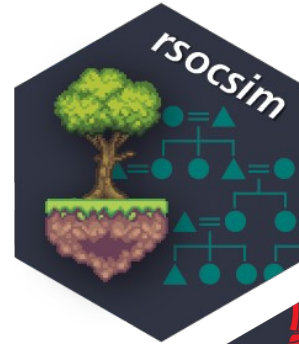
Reproducible Demographers

Diego
Alburez-Gutierrez

Liliana
Calderón-Bernal



High quality R libraries coming out of MPIDR



mpidr.github.io/rsocsim/



github.com/IvanWilli/DemoKin

and many more...

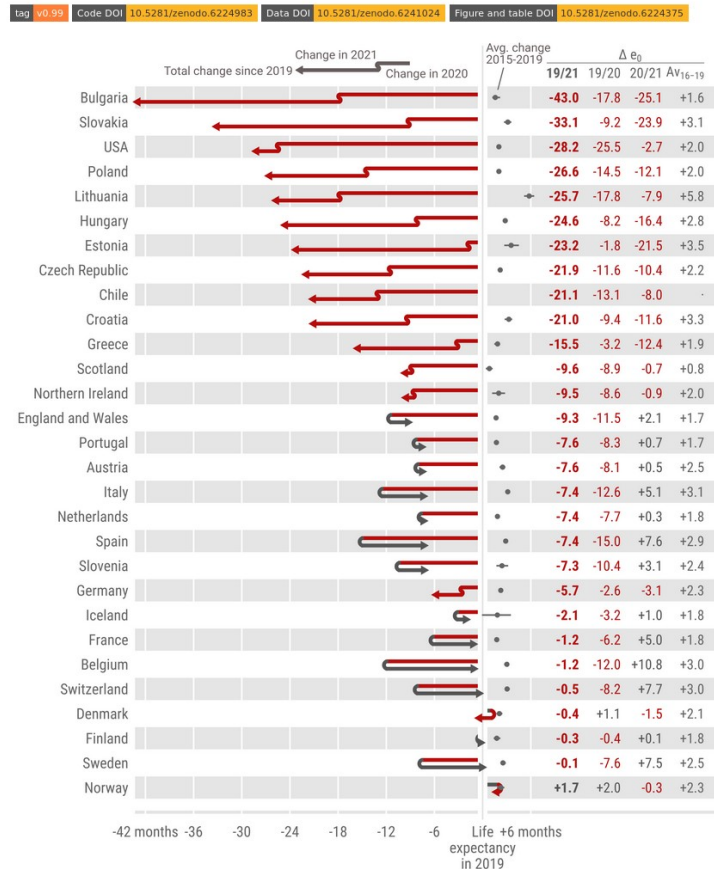


Tom **Theile**

Amanda Martins **de Almeida**

Consuming open science

Life expectancy changes since COVID-19



2.2 Data quality. Data: I see that all data were downloaded on February 18, 2022. Can you provide more information about the level of completeness or comparability of completeness across countries as of this download date? I am only familiar with the US data, which have a very long lag time for all-cause mortality and so any counts obtained are likely an underestimate as of this date. Even if they technically cover deaths through the end of 2021, many deaths get reported and processed at later dates. I would also recommend updating these estimates with the most recently available data when doing the next revision.

Consuming open science

 main ▾

 1 branch

 0 tags

Go to file

Code ▾

 **akarlinsky** Local Mortality Update d3a6d38 11 hours ago  **579** commits

	local_mortality	Local Mortality Update	11 hours ago
	preliminary_mortality	Preliminary Mortality update	15 days ago
	.gitignore	Update .gitignore	5 months ago
	LICENSE	Create LICENSE	11 months ago
	README.md	2022-06-07 Update	9 days ago
	coverage_map_title.png	Update coverage_map_title.png	2 months ago
	world_mort_plot_all.png	2022-06-10 Update	6 days ago
	world_mortality.csv	2022-06-10 Update	6 days ago

Karlinsky & Kobak (2022). World Mortality Database. github.com/akarlinsky/world_mortality

Consuming open science

The screenshot shows a GitHub repository page for 'akarlinsky Local Mortality Update'. The page is divided into two main sections: a file list on the left and a commit history on the right. A red box highlights the commit history, and a red line points from the '579 commits' badge to the commit history.

File List:

File Name	Last Update
local_mortality	Local Mortality Update
preliminary_mortality	Preliminary Mortality update
.gitignore	Update .gitignore
LICENSE	Create LICENSE
README.md	2022-06-07 Update
coverage_map_title.png	Update coverage_map_title.p
world_mort_plot_all.png	2022-06-10 Update
world_mortality.csv	2022-06-10 Update

Commit History:

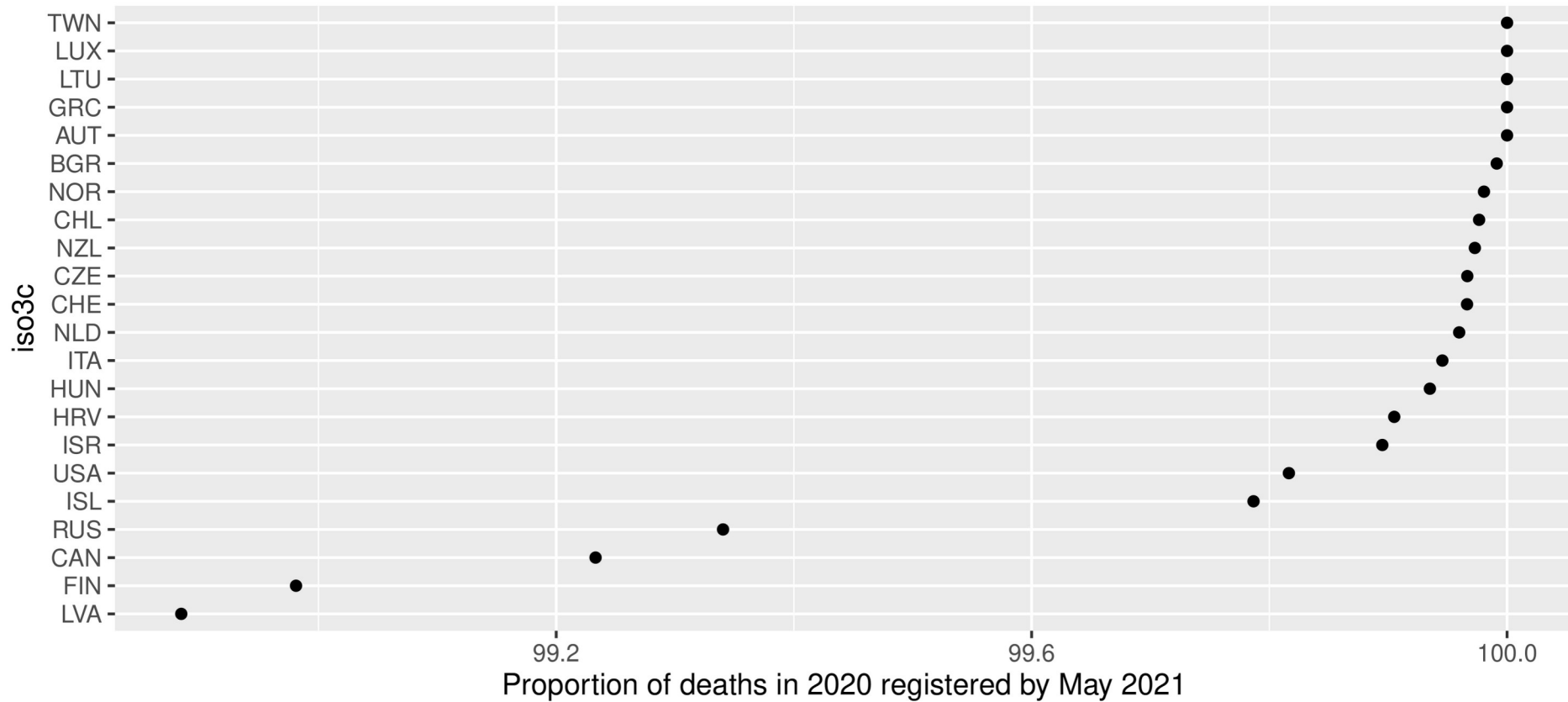
Commit Message	Author	Time Ago
2022-05-26 update	akarlinsky	committed 22 days ago
2022-05-24 update	akarlinsky	committed 24 days ago
2022-05-23 Update	akarlinsky	committed 24 days ago
Update world_mortality.csv	akarlinsky	committed 29 days ago
2022-05-19 Update	akarlinsky	committed 29 days ago
2022-05-15 Update	akarlinsky	committed on May 15

Repository Information:

- Repository: akarlinsky Local Mortality Update
- Branches: 1 branch
- Tags: 0 tags
- Commits: 579 commits

Karlinsky & Kobak (2022). World Mortality Database. github.com/akarlinsky/world_mortality

Consuming open science



Derived from Karlinsky & Kobak (2022). World Mortality Database. github.com/akarlinsky/world_mortality

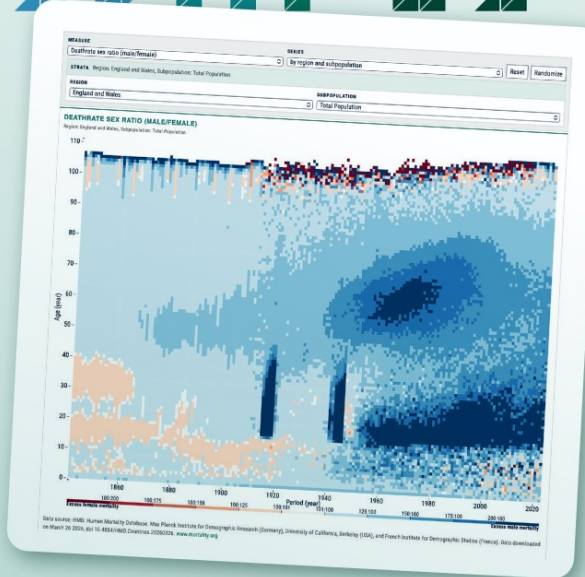
Open science as win-win



demoscapes.org

A visual atlas of demographic surfaces

Demoscapes.org is building a home for demographic surfaces, with interactive tools for seeing how population processes unfold across two time dimensions.



Join our community of contributors

Have your research featured by visiting
demoscapes.org/contribute.html
or by contacting **Jonas Schöley**
✉ schoeley@demogr.mpg.de



MAX PLANCK INSTITUTE
FOR DEMOGRAPHIC RESEARCH

Schöley (2026). [Demoscapes.org](https://demoscapes.org)

Given **your data** and **your analysis**
I arrive at **your results**

Given **your research question**,
my data and **my analysis**
I arrive at **your results**

4 + 1 Layers

Layer 0: Aspirational

Layer 1: Personal

Layer 2: Communal

Layer 3: Computational

Layer 4: Institutional

4 Layers

Layer 0: Aspirational

Realize you're having a problem



4 Layers

Layer 0: Aspirational

Layer 1: Personal

Layer 2: Communal

Layer 3: Computational

Layer 4: Institutional

Reproduce your own work

Project structure
Analysis pipeline
Documentation

4 Layers

Layer 0: Aspirational

Layer 1: Personal

Layer 2: Communal

Layer 3: Computational

Layer 4: Institutional

Let others reproduce your work



zenodo



4 Layers

Layer 0: Aspirational

Layer 1: Personal

Layer 2: Communal

Layer 3: Computational

Layer 4: Institutional

Automatize the reproduction of your work



4 Layers

Layer 0: Aspirational

Layer 1: Personal

Layer 2: Communal

Layer 3: Computational

Layer 4: Institutional

Embed reproducibility in your organization

Personal reproducibility

Communal reproducibility

Communal reproducibility

Ensure everyone
can run your
analysis



Share and version
your analysis



Share your data
Archive your data
Get DOIs



I'm not allowed to share my data

There is always a
Largest Shareable Derived Dataset

I'm not allowed to share my data

Identify your Largest Shareable Derived Dataset

Individual level data

Anonymized data subset

Model estimates

Data for plots and tables

Script 1: Subset and anonymize

Script 2: Fit model

Script 3: Create tables and plots

Example for sharing plot and table data. github.com/jschoeley/e0deficit.

I'm not allowed to share my data

Identify your Largest Shareable Derived Dataset

Individual level data

Aggregated data

Model estimates

Data for plots and tables

Script 1: Aggregate

Script 2: Fit model

Script 3: Create tables and plots

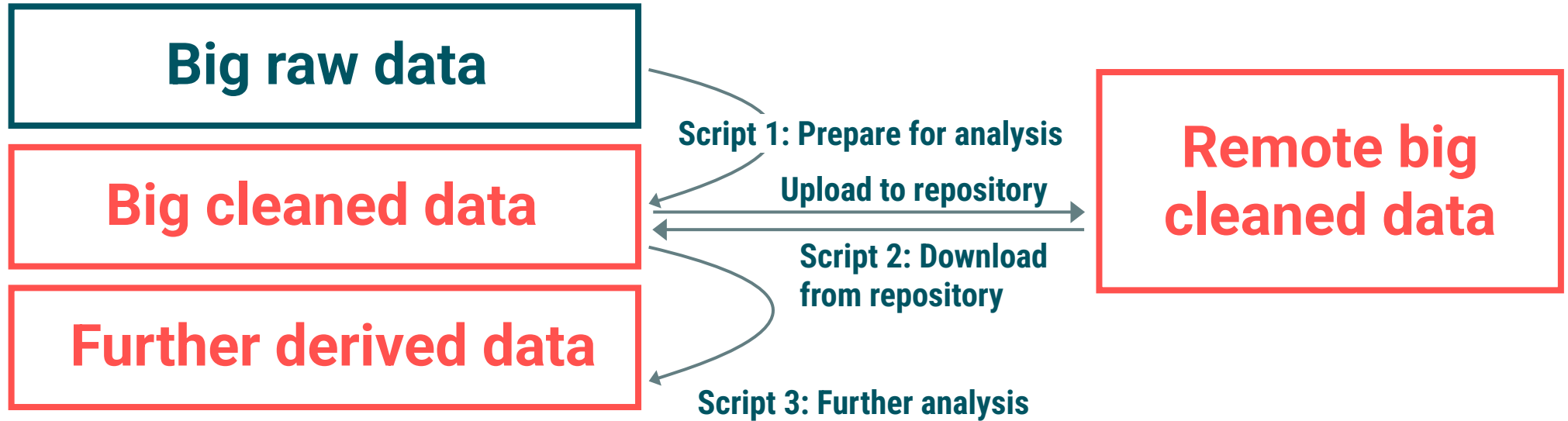
Example for sharing aggregated data. github.com/jschoeley/inselect.

My data is too big to share

Upload to a dedicated data repository
Data outsourcing

My data is too big to share

Upload to a dedicated data repository
Data outsourcing



Example for data outsourcing. github.com/jschoeley/inselect.

Computational reproducibility

The computational reproducibility stack

Ensure everyone
can run your
analysis

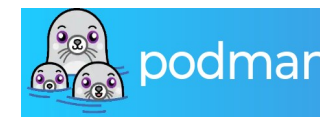


Scripts & Data

R Libraries

System dependencies

Hardware



The computational reproducibility stack

Ensure everyone
can run your
analysis

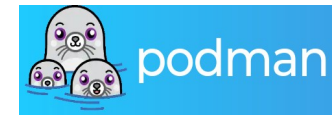


Scripts & Data

R Libraries

System dependencies

Hardware



The computational reproducibility stack

Ensure everyone
can run your
analysis



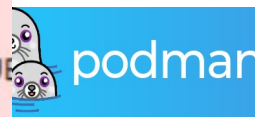
Scripts & Data

```
Warning message:
“`funs()` was deprecated in dplyr 0.8.0.
i Please use a list of either functions or lambdas:

# Simple named list: list(mean = mean, median = median)

# Auto named with `tibble::lst()`: tibble::lst(mean, median)

# Using lambdas list(~ mean(., trim = .2), ~ median(., na.rm = TRUE))
i The deprecated feature was likely used in the dataiku package.
Please report the issue to the authors.”
```



The computational reproducibility stack

Ensure everyone
can run your
analysis

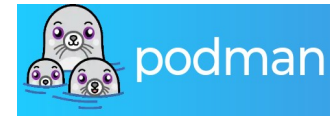


Scripts & Data

R Libraries

System dependencies

Hardware



The computational reproducibility stack

Ensure everyone
can run your
analysis

Scripts & Data



```
Error in stop_no_virtualenv_starter(version = version, python =  
python) :  
  Suitable Python installation for creating a venv not found.  
  Requested Python: /usr/bin/python3.10  
  Requested version constraint: 3.10  
Please install Python with one of following methods:  
- https://github.com/rstudio/python-builds/  
- reticulate::install_python(version = '<version>')  
- Install python3-venv and python3-pip using the system package  
anager
```



The computational reproducibility stack

Ensure everyone
can run your
analysis

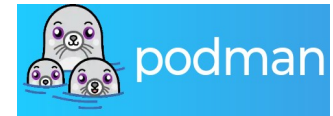


Scripts & Data

R Libraries

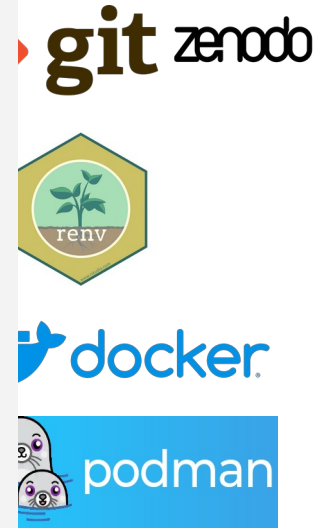
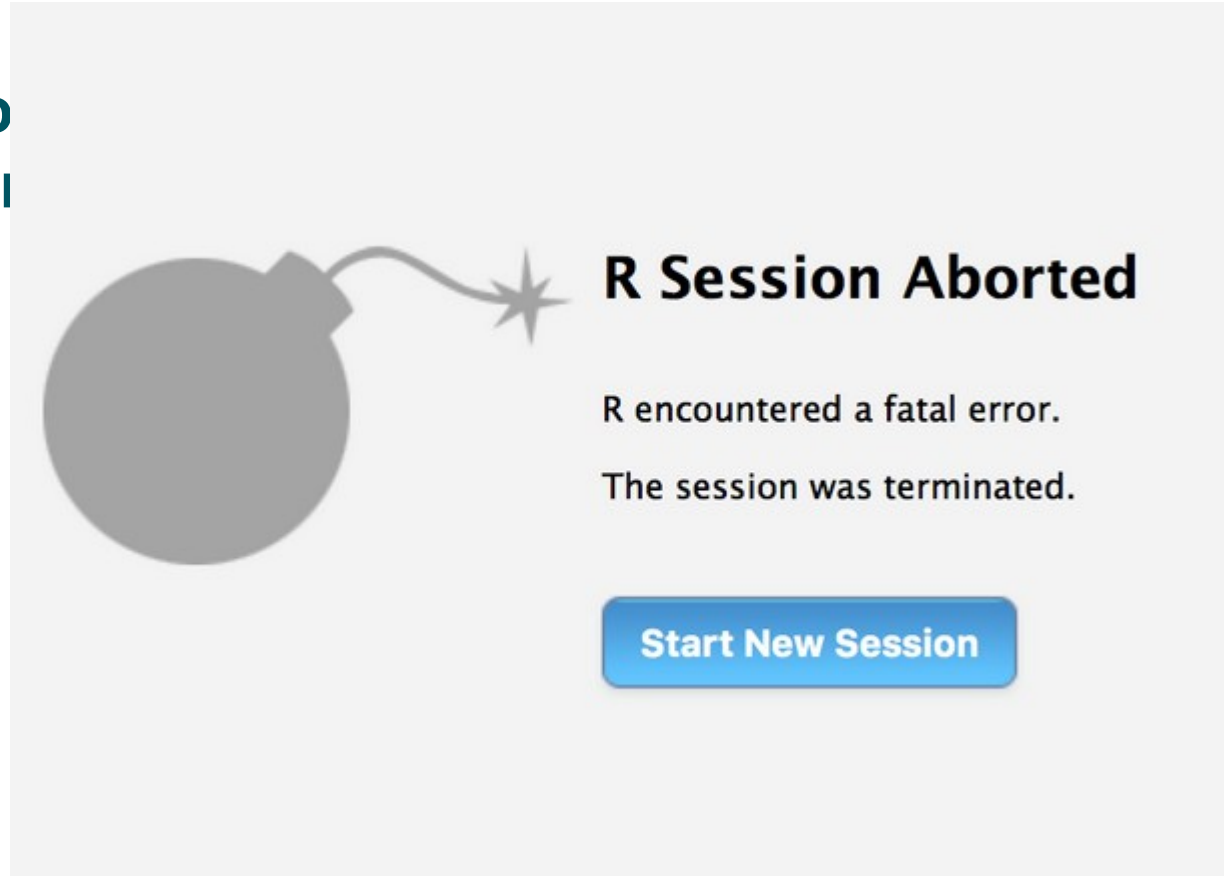
System dependencies

Hardware



The computational reproducibility stack

Ensure everyone
can run your
analysis



Institutional open science

Institutional open science



Aliakbar Akbaritabar



Open science at the Max Planck Society. osip.mpdl.mpg.de.

A few questions!

- Who **owns** the scripts/codes developed by researchers?
- Who should receive Open Science **training**?
 - At which **stage** of the **career**?
- Who should provide OS **infrastructure** and **maintain** it?
- What software solutions should be used and encouraged?
Proprietary?
- How should OS activities be **recognized/rewarded**?

What can institutions do for OS? (1/2)

- Permit and encourage researchers to **share** their research products under an **open license**.
- Particularly, researchers **with limited contracts** should have the right to access and publish all research products they developed during their time with the institute, even after they leave.
- **Train** researchers in OS practices, funded by the institute.
- Create a centralized, in-house **research data infrastructure**, and maintain it.
- **Binding guidelines** for the organization, storage, and retrieval of research data, ensuring continuity of access and management beyond the tenure of individual researchers or teams.

What can institutions do for OS? (2/2)

- Prioritize the use and development of **open-source research software**.
- Move away from **proprietary** software solutions.
- Micro-funds for **OS projects**, and/or **dedicated staff** to support OS.
- Transparent and reproducible research as a standard **criterion** for institute's **evaluation**.
- **Recognize** OS contributions by researchers and **reward** them, especially for **early career** researchers.

Hands-on part

What to expect in the Hands-on part?

- An interactive, self-paced, course on Git, GitHub, and version control with 4 steps:
 - Git and version control basics
 - GitHub - remote repositories and online collaboration
 - Branches, collaboration, and open science
 - Advanced Git, GitHub tools, and reproducibility
- **Requirement:**
 - Having a GitHub account, or opening one! Otherwise, follow on a neighbor's screen.
- **Link:**
 - <https://tinyurl.com/EPC260S>



Slides

github.com/jschoeley/epc26os

Hands-on materials

github.com/akbaritabar/Git-GitHub-Version-Control-Interactive-Self-paced-Course

Jonas Schöley

schoeley@demogr.mpg.de

 [@jschoeley.com](https://twitter.com/jschoeley)

jschoeley.com

Aliakbar Akbaritabar

akbaritabar@demogr.mpg.de

 [@akbaritabar.bsky.social](https://twitter.com/akbaritabar)

akbaritabar.github.io



MAX PLANCK INSTITUTE
FOR DEMOGRAPHIC RESEARCH